MANOJKUMAR MOHANKUMAR

Champaign, IL | mm240@illinois.edu | +1 (217) 841-0214 |
linkedin.com/in/manojkmohan5 | manojkmohan5.github.io/mkportfolio | github.com/manojkmohan5

SUMMARY

Software Engineer with 2 years of experience at Kyndryl building distributed backend systems for large-scale production workloads, currently pursuing a Master's at UIUC. Skilled in developing scalable cloud-native applications and AI-powered systems (RAG, LLMs), with a strong foundation in data structures, algorithms, and system design, focused on delivering reliable high-performance solutions.

EDUCATION

University of Illinois Urbana-Champaign **Aug 2025 – May 2027**
Master of Science in Information Management ***(GPA: 4/4)*** ***Champaign, IL***
Coursework: Data Warehousing and Business Intelligence, Data, Statistics and Information, Database Design and Prototyping

Anna University **Aug 2019 – May 2023**
Bachelor of Engineering in Computer Science and Engineering ***(GPA: 3.5/4)*** ***Chennai, India***
Coursework: Data Structures & Algorithms, Machine Learning Algorithms, Artificial Intelligence, Business Intelligence

- **Teaching Assistant** – Machine Learning Algorithms

SKILLS

- **Programming Languages:** Python, C/C++, JavaScript, TypeScript, SQL, Kotlin, JCL
- **Backend Systems:** REST APIs, Microservices, OOP, Distributed Systems, System Design, Concurrency, Threading
- **Frameworks & Tools:** Databricks, Django, Flask, FastAPI, MLFlow, Airflow, Kafka, Spark, Kubernetes, Power BI
- **AI & ML:** LLM APIs, Retrieval-Augmented Generation (RAG), Vector DB, Langchain, Semantic Caching, Ollama, Agentic AI
- **Cloud & Devops:** AWS (Textract, SageMaker, Bedrock, S3, EC2, ElastiCache, Lambda), Docker, CI/CD, Git, Firebase
- **Databases:** MySQL, PostgreSQL, MongoDB, Snowflake
- **Practices:** SDLC, Agile, Unit Testing, Debugging, Performance Optimization, Fault tolerance patterns
- **Certifications:** AWS Certified Cloud Practitioner (CLF-302), Microsoft Certified: Power BI Data Analyst Associate (PL-300), Google Associate Cloud Engineer, Stanford Machine Learning Online, AWS Partnered: Generative AI Essentials

PROFESSIONAL EXPERIENCE

Mainframe Systems Engineer – Kyndryl Solutions Pvt Ltd, Chennai, India **Sep 2023 – Aug 2025**

- Built distributed batch processing systems handling **1,000+ daily workloads**, ensuring high availability and SLA compliance.
- Designed automation services for job orchestration and failure recovery, improving operational efficiency by **25%.**
- Developed backend components integrated into **CI/CD** pipelines, reducing deployment downtime and improving release reliability.
- Reduced MTTR by **18%** through log analysis and debugging using **Python** and **SQL**.
- Improved system reliability by implementing fault-tolerant architectures and high-availability configurations.
- Led knowledge transfer and mentored **6+ engineers**, improving onboarding efficiency and reducing issue resolution time.

Data Science Intern – Exposys Data Labs, Bengaluru, India **Nov 2021 – Dec 2021**

- Built K-Means clustering model for customer segmentation, achieving **90%+ cluster stability** on transactional datasets.
- Performed feature engineering and preprocessing on time-series data to improve clustering performance.
- Evaluated model robustness through sensitivity analysis and improved interpretability of segmentation outputs.

Web Development Intern – The Sparks Foundation, Remote **May 2021 – Jun 2021**

- Built full-stack donation platform with payment integration, handling secure transactions and automated workflows.
- Developed **REST APIs** and responsive UI using **React**, improving mobile usability and system performance.
- Collaborated with 3 developers using pair programming; achieved **98% unit test coverage.**

TECHNICAL PROJECTS

Semantic Caching Framework for Context-Aware Conversational AI *[Github]* **Sep 2025 – Oct 2025**

- Built semantic caching system using embedding similarity search, vector storage, and session context, reducing LLM latency by **40%.**
- Managed multi-turn conversational context, used **Git** for version control, and optimized AI performance to reduce redundant API calls.

Agentic RAG Chatbot for Document Intelligence *[Github]* **Sep 2025 – Dec 2025**

- Built agentic RAG system using LangChain, vector databases, and tool orchestration, reducing response time by **35%.**
- Improved retrieval accuracy by **40%** through optimized embedding pipelines and similarity search.
- Implemented dynamic reasoning workflows using agent-based decision making (ReAct-style prompting).

LLM-Based Cyberbullying Detection (Human-in-the-Loop Evaluation) *[Github]* **Oct 2025 – Dec 2025**

- Built LLM-based classification system on **47K+ samples** using prompt engineering, supervised fine-tuning, and JSONL data pipelines.
- Designed human-in-the-loop (HITL) evaluation with A/B testing, improving macro-F1 by **18%** and reducing false positives by **22%.**
- Developed Trust & Safety workflow with confusion matrix analysis and feedback loops to improve model reliability.

RESEARCH PUBLICATIONS

- Published Research Paper, **"Adaptive ML Models for Real-Time Smart Grid Monitoring and Load Prediction**", *IEEE, Dec 2025*
- Published Research Paper, **"Interference Mitigation Using SDN Based Next Generation Networks"**, *ICRACSIT, Jul 2024*
- Published Research Paper, **"Artificial Neural Network Based Classification of Network Traffic",** *ICCSIT, Mar 2024*